

96.6

① a) Water

$$T = 327.86^\circ\text{F}$$

$$p = 100 \text{ lbf/in}^2$$

$$u = 500 \text{ Btu/lb}$$

$$x = \frac{u - u_f}{u_g - u_f} = \frac{500 - 298.3}{1105.8 - 298.3} = .25$$

$$h = h_f + .25(h_g - h_f)$$

$$h = 298.6 + .25(1187.5 - 298.6)$$

$$h = 298.6 + .25(1187.5 - 298.6)$$

$$h = 520.75 \text{ Btu/lb}$$

$$v = v_f + .25(v_g - v_f)$$

$$v = .01774 + .25(4.434 - .01774)$$

$$v = 1.121 \text{ ft}^3/\text{lb}$$

b) ammonia

$$T = 40^\circ\text{F}$$

$$x = .5$$

$$p = 75 - 70 = 5 \text{ lbf/in}^2$$

$$p = \frac{(41.11 - 37.67)}{(41.11 - 37.67)} (40 - 37.67) + 70 = 73.38 \text{ lbf/in}^2$$

$$p = \frac{5(2.33)}{3.44} + 70 = 73.38 \text{ lbf/in}^2$$

$$v_f = .02533$$

$$v_g = 3.9688$$

$$v = v_f + .5(v_g - v_f)$$

$$v = 1.99707 \text{ ft}^3/\text{lb}$$

$$u = u_f + .5(u_g - u_f)$$

$$u_f = 85.48$$

$$u_g = 568.42$$

$$u = 327.2 \text{ Btu/lb}$$

c) Water

$$p = 80 \text{ Bar}$$

$$v = .04034$$

$$T = 480^\circ\text{C}$$

$$u = 3025.7 \text{ kJ/kg}$$

$$h = 3348.4 \text{ kJ/kg}$$

d) Refrig

$$T = -20^\circ\text{C}$$

$$h = 235.31 \text{ kJ/kg}$$

$$p = 1.3299 \text{ Bar}$$

$$v = .1464 \text{ m}^3/\text{kg}$$

$$u = 215.84 \text{ kJ/kg}$$

no 2

2) Water
(S.V)

$$T_1 = 180^\circ\text{C}$$

$$T_2 = 100^\circ\text{C}$$

$$a) P_1 = 10.02 \text{ Bar} \times \frac{1 \text{ MPa}}{1 \text{ Bar}} = 1.002 \text{ MPa}$$

$$b) x = \frac{V_2 - V_f}{V_g - V_f} = 0.115$$

$$V_1 = 0.1941 \text{ m}^3/\text{kg} = V_2$$

3) Water (S.V)

(S.V)

$$m = 2 \text{ kg}$$

$$P_1 = 3 \text{ Bar} = P_2$$

$$V_2 = 2.064 \text{ m}^3$$

$$V_1 = 0.6058 \text{ m}^3/\text{kg}$$

a)

$$T = \frac{(140 - 130)(3 - 2.701)}{(3.613 - 2.701)} + 130 = 133.3^\circ\text{C}$$

$$b) V_2 = \frac{2.064}{2} = 1.032 \text{ m}^3/\text{kg}$$

$$V_2 > V_1 @ T_1$$

Superheated

$$P_2 = 3 \text{ Bar}$$

$$V_2 = 1.032 \text{ m}^3/\text{kg}$$

$$T_2 = 133.55^\circ\text{C}$$

$$c) W = P \Delta V$$

$$W = 3(2 \times 1.032 - 2 \times 0.6058)$$

$$W = 2.5572 \text{ kJ}$$

$$W = 3(2 \times (1.032) - 2 \times (0.6058))$$

$$W = 3(2.064 - 1.2116)$$

$$W = 2.5572 \text{ kJ}$$

4) Refrig 134a

$$P_1 = 10 \text{ Bar}$$

$$m = 1 \text{ kg}$$

$$T_1 = 60^\circ\text{C}$$

$$W = 20 \text{ kJ}$$

$$P_2 = 2 \text{ Bar}$$

$$T_2 = 30^\circ\text{C}$$

$$a) \text{ from table } V = 0.02301 \text{ m}^3/\text{kg} \times 1 \text{ kg} = 0.02301 \text{ m}^3 = V_1$$

$$b) u_1 = 268.35$$

$$u_2 = 253.06$$

$$Q = W + m(u_2 - u_1)$$

$$Q = 20 + 1(253.06 - 268.35)$$

$$Q = 4.71 \text{ kJ}$$

5) Water

$$T_1 = 70^\circ\text{C} \quad m = 1.5$$

$$T_2 = 120^\circ\text{C}$$

$$a) P_1 = \frac{(.4 - .3)(70 - 69.10)}{(75.87 - 69.10)} + .3 = .313 \text{ Bar} \times \frac{100.0 \text{ kPa}}{1 \text{ Bar}} = \boxed{31.33 \text{ kPa}}$$

$$b) \cancel{x} V_g @ 120 = \underbrace{V_f + x V_g}_{@ 70} \Rightarrow x = .17$$

@ T_1

$$h = h_f + .17(h_g - h_f) = \cancel{689.75} \text{ kJ/kg} \quad 689.75 \text{ kJ/kg}$$

$$h_f = \cancel{292.999} \text{ kJ/kg} \quad 292.999 \text{ kJ/kg}$$

$$h_g = \cancel{2626.83} \text{ kJ/kg} \quad 2626.83 \text{ kJ/kg}$$

@ T_2

$$h = 2706 \text{ kJ/kg}$$

$$h \times 1.5$$

$$h @ T_1 = 1034.63 \text{ kJ}$$

$$\boxed{Q = \Delta h = 3024.37 \text{ kJ}}$$

$$h @ T_2 = 4059 \text{ kJ}$$

6) Refrig

(S.L.V.)

$$P_1 = 10 \text{ Bar} \quad m_f = 25 \text{ kg}$$

$$x = .6$$

a) @ 10 Bar

$$V_f = .000835 \text{ m}^3/\text{kg}$$

$$V_g = \cancel{.000835} \text{ m}^3/\text{kg} \quad .0236 \text{ m}^3/\text{kg}$$

$$V = V_f + x(V_g - V_f) = .0145 \text{ m}^3/\text{kg}$$

$$\cancel{x} = \frac{m_g}{m_g + m_f} = \boxed{37.5 \text{ kg} = m_g} \quad \leftarrow \text{error circling}$$

Vol of tank

$$V = m \times V = \cancel{25 \times .000835} = (25 + 37.5) \times .0145 = \boxed{.9063 \text{ m}^3}$$

$$b) V_g = .0236 \text{ m}^3/\text{kg} \times 37.5 \text{ kg} = .885 \text{ m}^3$$

$$V_f = .000835 \times 25 = .0209 \text{ m}^3$$

$$\% \text{ Vap} = \frac{.885}{.9063} = \boxed{97.65\%}$$

$$7) m_h = .2 \text{ g} \Rightarrow .0002 \text{ kg}$$

$$P_1 = 1.3 \text{ Bar} \Rightarrow 130 \text{ kPa}$$

$$T_1 = 27^\circ\text{C} \Rightarrow 300^\circ\text{K}$$

$$P_2 = .8 \text{ Bar} \Rightarrow 80 \text{ kPa}$$

$$T_2 = 17^\circ\text{C} \Rightarrow 290^\circ\text{K}$$

$$\bar{R} = 8.314 \text{ kJ/kmol}\cdot\text{K}$$

$$a) R_h = \frac{\bar{R}}{m} = \frac{8.314}{4} = 2.0785 \text{ kJ/kg}\cdot\text{K}$$

$$V_1 = \frac{m R_h T_1}{P_1} = \frac{.0002 \times 2.0785 \times 300}{130} = .000959 \text{ m}^3 \Rightarrow 959 \text{ cm}^3$$

$$V_2 = \frac{m R_h T_2}{P_2} = \frac{.0002 \times 2.0785 \times 290}{80} = .001506 \text{ m}^3 \Rightarrow 1506 \text{ cm}^3$$

$$8) T_1 = 25^\circ\text{C} \Rightarrow 298^\circ\text{K}$$

$$P_1 = 2.75 \text{ Bar} \Rightarrow .275 \text{ MPa}$$

$$V_1 = .05 \text{ m}^3$$

$$T_2 = 257^\circ\text{C} \Rightarrow 530^\circ\text{K}$$

$$Q = -2.75 \text{ kJ}$$

$$R = .189 \text{ kJ/kg}\cdot\text{K} \Rightarrow 189 \text{ J/kg}\cdot\text{K}$$

$$a) P_1 V_1 = m R T_1$$

$$m = \frac{P_1 V_1}{R T_1} = \frac{.275000 \text{ Pa} \times .05 \text{ m}^3}{189 \text{ J/kg}\cdot\text{K} \cdot 298 \text{ K}} = .244 \text{ kg}$$

$$b) Q = U + W$$

$$Q = n(u_2 - u_1) + W$$

$$n = \frac{.244 \text{ kg}}{.04401 \frac{\text{kg}}{\text{mol}}} = 5.544 \text{ mol}$$

$$u_1 = 6.90 \text{ kJ/mol}$$

$$u_2 = 14.70 \text{ kJ/mol}$$

$$W = -2.75 - 5.544(14.7 - 6.9) = -45.993 \text{ kJ}$$